LAB : 11

Title : Strings in C++

► Objectives

- To understand C-Strings (character arrays terminated with \0).
- To learn declaration and initialization of C-Strings.
- To explore String objects in C++.
- To apply string functions for manipulation (length, concatenation, comparison, etc.).

► Theory

1. C-Strings

- A C-String is an array of characters ending with a null character \0.

Example:

```
char myString[] = "Hello";
```

2.String Objects

· he string class in C++ (from `<string>` header) allows dynamic length strings.

· Input can be taken using `getline(cin, str)` which reads entire lines including spaces.

3. String Functions

Common C-String functions include:

- `strlen(s)` → length of string
- `strcpy(s1, s2)` → copy string
- `strcat(s1, s2)` → concatenate strings
- `strcmp(s1, s2)` → compare strings
- `strrev(s)` → reverse string
- `strlwr(s)` → convert to lowercase
- `strupr(s)` → convert to uppercase

► Lab Tasks

Task 1: String Length Calculation

```
#include <iostream>          // tells the compiler to include input output strem
library

#include <cstring>           // include string functions

using namespace std;          // tells the compiler to use standard namespace



int main() {                                        // main function begins from here

   char message[] = "Electrical Engineering";         // declare and initialize string

   cout << "Message: " << message << endl;                    // display the string

  cout << "Length of message: " << strlen(message) << endl; // calculate and show
length

   return 0;                                                 // end program

}
```

Task 2: String Concatenation

```cpp
#include <iostream>                          // include input output stream library
#include <cstring>                           // include string functions
using namespace std;                         // use standard namespace std


int main() {                                 // main duction starting point
    char firstName[20] = "Naeem";            // first name
    char lastName[20] = " ul Haq";           // last name
    strcat(firstName, " ");                  // add space after first name
    strcat(firstName, lastName);             // join last name to first name
    cout << "Full Name: " << firstName << endl;    // display full name
    return 0;                                // end program
}
```

► Conclusion

In this lab, we explored both C-Strings and String objects in C++. C-Strings are fixed-size character arrays terminated with a null character, while String objects provide more flexibility with dynamic length and easier handling of spaces. We practiced using built-in functions such as `strlen()` for measuring string length and `strcat()` for concatenation, which demonstrated how text data can be manipulated efficiently.

The exercises showed the limitations of `cin >>` (stopping at spaces) and the advantages of `cin.get()` and `getline()` for reading complete lines. Overall, this lab strengthened our understanding of how strings are stored in memory, how they can be processed, and why modern C++ string objects are often preferred for real-world applications.